Rohit Acharya

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Education

National Institute of Technology (NIT) Rourkela Odisha, India
*B.Tech. in Civil Engineering — CGPA: **7.96 / 10.0*** *Aug 2021 – May 2025*

Thesis: *Buckling Performance Evaluation of FG-GRC Plates under Thermal-Mechanical Loads* — Prof. Shishir Kumar Sahu

Courses: FEM, Advanced Structural Analysis, Mechanics of Solids, RC Design, Foundation Engineering, Fluid Mechanics, Numerical Analysis

Kathmandu BernHardt Secondary School Kathmandu, Nepal
*Higher Secondary, Science — CGPA: **3.85 / 4.0 (NEB)*** *2018 – 2020*

Research Experience

Undergraduate Researcher — Computational Structural Mechanics Jun – Dec 2024
Dept. Civil Engineering, NIT Rourkela Supervisor: Prof. Shishir Kumar Sahu *Rourkela, India*

- **Formulation:** Modeled buckling of FG-GRC plates using Classical Plate Theory (CPT); solved eigenvalue problem $[\mathbf{K}]\{u\} = \lambda[\mathbf{K_G}]\{u\}$; derived material gradation via power-law distribution and Halpin–Tsai micromechanical model
- **Simulation:** Parametric FEA in ABAQUS across 12+ variables — gradient index n, aspect ratio a/h, GNP fraction V_{g0} (5–25%), thickness (2–10 mm), temperature (300–500 K), boundary conditions (SSSS, CCCC, SSCC)
- **Key Findings:** 25% GNP fraction $\Rightarrow$ ≈30% higher critical buckling load; 500 K thermal rise $\Rightarrow$ ≈15% degradation; CCCC sustained ≈20% more post-buckling load than SSSS
- **Validation:** Mesh convergence at 4 mm (C3D20R elements); imperfection sensitivity via sinusoidal perturbation; post-buckling via nonlinear static analysis

Seminar Research — FRP Strengthening of RC Structures 2024
Dept. Civil Engineering, NIT Rourkela CE4900: Seminar & Technical Writing *Rourkela, India*

- Critically reviewed CFRP/GFRP/BFRP literature (Askar et al. 2022; Mofidi & Chaallal 2014; Attari et al. 2012); analysed EB vs. NSM systems for flexural, shear, and axial RC strengthening
- Derived and validated formulations for $M_{u,FRP}$, v_{FRP} (U-wrap), and f'_{cc} (FRP jacketing); confirmed 60–85% flexural, 200% shear, 50–60% axial gains; contextualised for Indian seismic zones

Professional Experience

CloudFactory Kathmandu, Nepal
Construction Analyst (Buildots — AI Construction Progress Tracking) *Jul 2025 – Present*

- Verify construction progress from 360° walkthrough data against BIM models and schedules; systematic QA audits on annotated construction datasets; identify sequencing anomalies and deviations
- Contribute to internal annotation standards and quality frameworks used to train AI construction progress models

Envirocare Infrastructure Pvt. Ltd. (Shri Mahavir Ferro Alloys) Rourkela, India
Project Intern — Site Execution & Estimation *Jan – May 2025*

- Supervised civil works (New SMS, Rolling Mill, Pump House); prepared BOQ and RA bills from site measurements and layout drawings; coordinated contractors, site engineers, and billing teams

Bureau of Indian Standards (BIS) Guwahati, India
Intern — Quality Assurance, Civil Engineering Division *May – Jul 2024*

- Applied IS codes to material compliance testing (cement, aggregates, steel); assisted in document audits and analysis of compliance reports

Projects

Structural Health Monitoring Dashboard | *Live Demo* | Next.js · Python · SciPy · Time-Series 2025 – Present

- Web-based SHM platform ingesting accelerometer and strain gauge data; computes FFT features and modal parameters; flags anomalies via statistical thresholds — designed for bridge and building monitoring

FG-GRC Plate Buckling Simulator | *GitHub* | Python · NumPy · Streamlit 2025

- Parametric buckling calculator implementing Halpin–Tsai model and CPT eigenvalue solver; outputs critical load vs. n, V_{g0}, temperature, aspect ratio; validated against ABAQUS thesis data

RC Structure FRP Retrofit Analyzer | *GitHub* | Python · Streamlit 2025

- Computes $M_{u,FRP}$, v_{FRP}, f'_{cc} gains for user-defined RC sections and CFRP/GFRP configs; generates load-deflection and stress-strain comparison plots with IS-code-aligned inputs

B.Tech. Thesis — FG-GRC Plate Buckling Study | *Report* | ABAQUS · MATLAB · Python 2024

- Full parametric FEA (12+ variables); 25% GNP $\Rightarrow$ ≈30% buckling load gain; 500 K $\Rightarrow$ ≈15% degradation; CCCC $\Rightarrow$ ≈20% higher post-buckling resistance than SSSS

Skills

Simulation / FEA: ABAQUS (eigenvalue buckling, nonlinear static, post-buckling, C3D20R), MATLAB
Programming: Python (NumPy, SciPy, Matplotlib, Pandas, Streamlit), MATLAB scripting, JavaScript / Next.js
Civil / Structural: AutoCAD, STAAD.Pro, BIM concepts, BOQ & estimation, construction scheduling
Data & Platforms: MS Excel (advanced), data annotation & QA, Buildots (AI construction platform), time-series analytics
Languages: English (Professional/C1), Nepali (Native), Hindi (Fluent)

Awards & Achievements

CGPA 8.20 at end of Semester 4 — NIT Rourkela
Grade-10 GPA 3.95 / 4.0, Outstanding Distinction — Mathematics, Science, CS; NEB Nepal (2018)
Grade-12 CGPA 3.85 / 4.0 — A+ in Physics, Chemistry, Mathematics; NEB Nepal (2020)
Aspire Institute Leadership Program (2024) — Competitive global cohort; leadership & cross-cultural collaboration

Leadership & Service

Student Mentor May 2023 – May 2025
Institute Counselling Services, NIT Rourkela

- Academic, personal, and career guidance for junior students in NIT Rourkela's structured peer-support programme

Founder & Lead Educator Jul 2020 – Jun 2024
Learners Club, Kathmandu, Nepal (YouTube)

- Free STEM education platform during COVID-19; Physics, Chemistry, Mathematics; 20,000+ views; led a volunteer team of student educators

Program Coordinator | Student Member (ASCE) Jun 2022 – May 2025
SDG Campus Club, NIT Rourkela | American Society of Civil Engineers

Certifications

Python for Data Science, AI & Development — IBM / Coursera (2024) | Foundations: Data, Data, Everywhere — Google / Coursera (2024) | Introduction to Thermodynamics — Coursera (2023) | AWS S3 Basics — AWS / Coursera (2023)